

Applications of Artificial Intelligence

Unit 2 Consolidate - Peer Instruction and Group Tasks

Answers

Group Task 1:

AI is the pursuit and engineering of systems that can stimulate human intelligence and thought processes.

(stimulate, in this definition emphasises that AI systems use machine learning, and predictive mechanisms to deliver outputs that emulate human thinking, rather than go through the same, creative thinking processes that humans go through).

Group task 2:

The Turing Test: A human evaluator asks questions that will then be answered by both another human and a machine in natural language.

The machine passes the test if the evaluator cannot tell the difference between the human and computer's answers.

Pro Argument: "The Turing Test is a valid benchmark for AI intelligence"

- The test measures the ability of AI to produce answers that are similar to human answers. While not a perfect benchmark, it has been a relevant consideration in the design and evaluation of AI systems and Natural Language Processing (NLP).
- The test focuses on observable behaviour, rather than internal processing, which makes it simple and easy to understand.

Con Argument: "The Turing Test is outdated and irrelevant today"

- The Turing test focuses on the ability of AI to produce output that imitates humans, rather than the accuracy of said output.
- The test relies on the judgement of one human, the evaluator, which is subjective.

Group Task 3

Case Study Analysis of a chosen AI Breakthrough

The DeepBlue vs Kasparov (Chess Computer, 1996-1997)

This was a customised chess-playing computer, which beat Garry Kasparov, the world champion in chess at the time, in an arranged match.

The importance of the breakthrough:

- It demonstrated the capabilities of AI-powered systems as an opponent in chess, which is linked to its capacities as an opponent in other games, including many modern video games.

Chess-playing AI systems have since improved. They are able to function much faster than in the 1990s and can play at a wide range of abilities, making them a great teaching and learning aid.

Most notably, there have been tournaments in recent years pitting AI systems against each other to examine the system's abilities for reasoning and strategy, notably a 2025 tournament between Grok and Open AI.

(<https://www.bbc.co.uk/news/articles/ce830l92p68o>)